

Visual art history and the psychology of perception: Perspectivism and its 20th century abandonment in the visual arts and in Gibson's ecological psychology

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Abstract

The development of linear perspective in the early 15th century and the discovery of the retinal image two centuries later became cornerstones of an approach to visual perception theory that eventually took shape primarily in the hands of British Empiricist philosophers. Even as this approach has dominated perceptual theory to the present day, the perspectivist influence on pictorial representation within the visual arts steadily diminished over time. Its decisive break with perspectivism came in the early 20th century with transformative 19th century changes in the sciences and technology. Collectively, these events elevated process and change over fixity and stasis, and ultimately led to the collapse of the distinction between space and time in the physical sciences. Even so, approaches to visual perception in psychology remained remarkably untouched by these occurrences until the 1960s when the experimental psychologist James Gibson drew upon them to challenge the legacy of perspectivism and the visual image and their effect on perceptual theory. His ecological approach to perception recognizes animacy as the essential functional property of living things, and in doing so, conceptualizes seeing as a perception–action process. From this stance, Gibson like the visual artists earlier in the century rejected the assumption that visual perception is

best characterized as the capturing of static images. Jointly and yet independently, both efforts loosened the grip that perspectivism and the visual image have maintained on the arts and on visual perception theory, respectively, bringing 19th century scientific advances into 20th century psychological thought.

KEYWORDS

ecological psychology, history of psychology, perspective, picture perception, visual perception

1 | INTRODUCTION

Seeing as such has its own history. (Wofflin, 2015, p. 93)

The history of theories of visual perception is a well-trod ground, from Boring's (1942) seminal book, to later important contributions including Hamlyn (1961), Pastore (1971), and Hatfield (1990), among others. This is a history not solely for the benefit of historians, however. It is essential ground for psychologists working within nearly any area of the discipline because assumptions as to the nature of visual perception, broadly considered, underwrite most contemporary psychological theory. In fact, the relevance of theories of visual perception for psychological theory generally has taken on special significance since the "cognitive turn" in experimental psychology of the 1960s.

As cognition became a central focus of experimental psychology at that time—and in doing so, moved away from a more functional behaviorist orientation—the received views of visual perception *were uncritically taken up* as the foundation for cognitive theories across a wide range of domains. Their assumptions about visual perception have shaped current thought and have a bearing on continuing trends in psychology, epistemology, and allied areas of inquiry. For this reason, the historical roots of theories of visual perception merit the scrutiny of psychologists beyond those specifically interested in perception and its history.

The consensus among historians of perception such as those cited above is that the approach to visual perception that came to dominate modern experimental psychology took shape by at least the 17th and 18th centuries within the Newtonian scientific and cultural climate of that time. Even so, the roots of this approach are traceable to developments centuries earlier than that, with the creation of linear perspective techniques in pictorial representation at the time of the Renaissance, which a few centuries later converged with findings about the structure of the eye and the "discovery" of a retinal image. The legacy of this history is the assumption that visual perception begins with the reception of *static images*. Contemporary perception theory retains many features of these origins, as we will see.

The fact is, however, that a great deal has changed culturally and scientifically since the 17th and 18th centuries, particularly over the course of the 19th century. Indeed, during that fertile period there was a wholesale transformation in our scientific and cultural outlook (*Weltanschauung*), with evolutionary theory in biology, field theory in physics, and technological changes among the developments driving it. In short order, these developments set the stage for artistic challenges to traditions of pictorial representation, eventually resulting at the outset of the 20th century in the abandonment of linear perspective by painters attempting to integrate space and time on a two-dimensional surface. These 19th century developments also began to transform received views of perception within psychology and philosophy, initially through the writings of William James and John Dewey among others,

who argued for an emphasis on process and change in psychological theory. Working within their pragmatist tradition, James Gibson began to formulate by the 1960s a through-going reconceptualization of perception—the ecological approach—that abandoned the pre-19th century assumptions that have sustained mainstream theories of visual perception to the present day. By recognizing animacy as an essential property of living things, Gibson conceptualized seeing as a perception–action process that reveals structure in the environment over time. In this latter respect, the distinction between space and time collapses in the realm of perception as it previously did in the visual arts—and both against a background of dynamic field theory in physics and the alterations of *everyday experience* that resulted from technological change.

This paper will begin by tracing the early interconnections between developments in pictorial representation and studies in optics, which when later folded into a Newtonian worldview, shaped modern theories of visual perception. Following that, the transformative effects of 19th century developments in science and technology on pictorial representation will be reviewed, as well as their contribution to the emergence of Gibson's ecological approach to perception.

2 | TWO ENDURING ASSUMPTIONS ABOUT VISUAL PERCEPTION: EXPLICATING THE DOGMA

In spite of the wide variety of approaches to visual perception proposed since the beginnings of modern science at the time of the Renaissance, two related assumptions about visual perception have endured over time and are cornerstones of the received view: (a) What is delivered immediately to perceivers is a static array of light on the two-dimensional surface of the retina and (b) perceiving individuals receive this projected array of light at a stationary observation point. Underlying both of these assumptions is a claim that be traced back to antiquity: Visual perception is based on the reception of images.

The preceding two assumptions established the agenda for much of perception literature by ushering in several core problems that exist *principally as a result of accepting those assumptions*. Here, we name three of them: (1) If the immediate display of light on our retina is lacking in depth/distance from a stationary point of view, how are we to account for the fact that our perceptual experience is of a world that extends away from us spatially in all directions? (2) Because the perceived size of an object is *correlated* with its perceived distance, if distance is not immediately available in the array of light projected on the retina at a single observation point, how do we explain the perceived absolute and relative size of objects? (3) Because an object's perceived shape is by definition a three-dimensional object property, if the immediate projected array of light is static and lacking in depth, then how do we account for our experience of object shape as opposed to merely object form?

What is the basis for these two taken-for-granted assumptions that have long been mainstays of inquiry into visual perception? The answer to this question can be found in the history of efforts to understand the nature of vision. Locating these assumptions in the history of visual perception theory, and particularly as they arise in relation to developments in the visual arts, will allow us to gain insight into their origins. Ultimately, doing so will also afford us an opportunity to consider whether these are necessary assumptions for an adequate account of visual perception or historically contingent ones.

3 | PERCEIVING “IMAGES”

There is a long tradition going back to antiquity that vision is based on obtaining “images” (likenesses or *eidola*) of things in the world (Lindberg, 1976, Ch. 1). Before the 12th century, there were two competing accounts of how these images are obtained. Discussions of these accounts appear in numerous sources. Here, I rely heavily on Lindberg's (1976) now classic book and the more recent work by Smith (2015).

The “extramission” approach posited that light rays emanate from the eyes, extending into the world and make contact with the objects of visual perception. By most accounts, this approach can be found initially in the writings of Euclid (fl. 300 b.c.) who proposed that rectilinear rays are emitted from the eye and travel along diverging paths making contact with an object's surface. These diverging rays form a *visual pyramid* or cone, with its vertex at the eye and, importantly for our purposes here, its *base* on the surface of the object (Lindberg, 1976; see Smith, 2015, Ch. 2, for a detailed treatment of Euclid's optics).

Euclid exemplifies those theorists whose primary concern was a geometric treatment of optics rather than on the physical effect of light on the eye or psychological treatment of vision. It has been suggested that Euclid's treatment of optics was intended primarily as a tool for use in scenography (i.e., design of backgrounds for theatrical performances) rather than as an explanation of vision (Hatfield & Epstein, 1979, footnote, 11; also see Lindberg, 1976, pp. 219–220, footnote 72; also see, Wartofsky, 1980 on scenic design). The images painted on the surfaces used for theater would be captured by visual rays emitted from the eye of a viewer.

Ptolemy (fl. 127–148 AD) subsequently and consequentially adapted what may have been primarily an abstract geometric analysis for scenic design to an extramission account of visual perception of objects in the world. Because the base of the visual cone positioned as it is on the surface of an object in the world is two-dimensional by definition, the perceived size and shape of an object both which also require distance information (as we saw above) cannot be explained only with reference to the object properties displayed at the base. Ptolemy proposed that perceived distance is to be attributed to the “felt length” of the light rays coming from the eyes when they make contact with a materially compact surface (Smith, 2015, p. 83). The felt length and the visual angle subtended by converging light rays to the surface would jointly give rise to perceived size of an object, and perceived shape was due to the form and its orientation detected through felt distance on the base of the visual cone (see, Smith, 2015, fig. 3.1).

The distinction mentioned above between a largely geometric analysis of vision and a psychological one is apparent as Ptolemy proposes that the rays of light are to be treated for the purposes of geometrical analysis as virtual or mathematical abstractions rather than real. In reality, visual radiation is a *flux*, “absolutely continuous in all dimensions, not actually composed of discrete visual rays” (Smith, 2015, p. 80). Ptolemy makes this distinction because the notion of discrete rays of light that lend themselves to geometric analysis is inconsistent with the perceiver's *experience* of surfaces, which are continuous rather than pointillistic or fragmented. This distinction between the kind of abstract analysis that geometric optics offers and an account of how the world is experienced is well worth bearing in mind so that we do not lose sight of the perceptual phenomena to be explained (Heft, 2003). Ptolemy proposes that the flux of visual radiation makes contact with the surface of any object and its effects are “transmitted back to the eye, [with] the flux serving as a continuous *medium* to support that transmission” (Smith, 2015, pp. 82–83, emphasis added).

As intellectually intriguing as these efforts are, extramission proposals will immediately strike contemporary readers as fanciful in the extreme. Indeed, in due course they will be abandoned (see below). Still, they do accord, at least, with the phenomenology of the directed selectivity of vision. Colloquially speaking, individuals do seem to pick out “what” to look at, just as in haptics, they pick out what to touch. There does seem to be what will later be called intentionality in vision, and that aspect of seeing continues to be a challenge on the contemporary scene.

The competing “intromission” approach was based on the claim that objects shed *eidola* that cross the gap (the air) between object and perceiver, impressing themselves on the eye. According to Democritus (ca. 460 BC), a material facsimile of the object radiates from the object making contact with the eye. Aristotle's (384–322 BC) account follows this line of thinking except that a transformation of the air by the Form of the object (formal causality) rather than the transmitted matter (*eidola* as such) makes it possible for the visual sense to take up that Form resulting in its perception.

When a geometric analysis is later applied to the intromission account, light is proposed as radiating in rays in all directions from the *surfaces* of objects, with rays from adjacent points on an object's surface converging at the lens of the eye and retaining their adjacent relationships. Hence, the notion of a visual cone with its base on the

object perceived and its vertex at the eye that was held in an extramission approach is repeated here, but the source of radiating light rays shifted from the eye to the features of the world; and the two-dimensional arrangement of light at the base of the visual pyramid is copied onto the surface of the lens, which at this juncture is a cross-section of the visual pyramid. As is the case with extramission proposals, this approach relies on the analytical tools provided by Euclidean geometry.

Owing primarily to Arab scholars who preserved as well as advanced Greek thought, and notably to the work of Alhazen (965–1039), geometric (perspectivist) studies of vision moved away from a nearly exclusive abstract analysis to include biological and functional considerations (see Lindberg, 1976, Ch. 2–4). Alhazen's writings on optics and vision, which were translated for Western scholars by the early 13th century, established the intromission approach to visual perception that was to be followed in subsequent centuries. Extramission approaches were largely abandoned from this point on.

Alhazen maintained that the light rays radiating from objects and that converge in the eye encounter the anterior surface of the lens, (i.e., the anterior glacial humor). As described above, there a point-by-point projection of light rays is displayed corresponding to their arrangement on the base of the posited visual cone at the object's surface. The anterior glacial humor is positioned as a cross-section of the visual cone where the light rays fall "in precisely the same order as the points in the visual field from which they originated" (Lindberg, 1976, p. 74). It was generally assumed that what was conveyed by the rays was "luminosity and color of each ray—and not the arrangement of the rays or the spatial information conveyed by that arrangement" (Hatfield & Epstein, 1979, p. 367). That arrangement of the rays on the lens was ultimately received or taken up by the faculty of sense which was positioned beyond the optic chiasm (the juncture of the optic nerves from each eye). The most important point for our purposes is that what was immediately received on a light sensitive surface of the eye was lacking in depth, with depth to be added postocularly.

This overall conceptualization of optics and the initial phases of visual perception was maintained over the next several centuries with modest modification in the hands of numerous scholars working in the tradition of Alhazen (although I skirt numerous debates during this period, see Lindberg, 1976). This viewpoint prevailed until the early 17th century when Kepler (1571–1630) reported finding an image projected on the rear interior surface of the eye (the retina), and proposed an analysis of the intraocular optics involved in producing this image. With this discovery, what was taken to be the initial seat of vision, recessed from the surface of the lens to the retina.

Kepler is best known in the natural sciences for his astronomical research that established the elliptical orbits of the planets. In the course of refining measurements in that work, he was led to investigate the ways in which light was affected when it passed through an aperture (Lindberg, 1976; Straker, 1981). Some of this latter work involved the use of a *camera obscura* or pinhole camera. The *camera obscura*, which has roots in Antiquity, was already quite familiar to scholars by the late Middle Ages. When light passes through a narrow aperture in a surface of an otherwise enclosed, dark chamber, an *image* of the area in front of the aperture is projected on the chamber's interior wall opposite to the aperture. That image can be sharpened through the use of a lens. The camera obscura was a forerunner of the modern camera (see below).

The workings of the eye had been directly compared to that of a camera obscura by Leonardo in the late 1400s and by Della Porta in the 16th century (Lindberg, 1976). However, neither claimed that the retina functioned like that of the rear interior surface of the camera obscura as Kepler later would (Lindberg, 1976, p. 164). Still, according to Lindberg (1976), Kepler invoked the camera obscura solely as an analogy as to how vision worked. It was Kepler's careful anatomical research, and particularly his knowledge of the tradition of optics among medieval scholars, that led him to identify the retina as the site where vision is initiated, and that what was projected on the retinal surface was an image. However, Kepler introduced a new problem: that of the inverted retinal image compared to the appearance of the world before the eye (Stoffregen, 2013). Most important for our purposes here is that Kepler returned the issue of an *image* delivered to the eye as a central concern for a theory of visual perception, which had been somewhat overshadowed by concerns about geometric optics and debates about where in the eye visual perception begins.

Kepler proposed that each ray of light is refracted first by the cornea and then by the lens, resulting in a set of rays, each “a shorter and blunter cone... with its apex on the retina” (Lindberg, 1976, p. 200). Thus, there is an “an orderly one-to-one correspondence between the point sources of radiation [in the world] and points stimulated within the eye [at the retina]” (Lindberg, 1976, p. 193). In 1604 Kepler writes: “Therefore vision occurs through a *picture* of the visible thing on the white, concave surface of the retina...” (quoted in Lindberg, 1976, p. 200, emphasis added). Lindberg points out that Kepler's use of the term *picture* to refer to this display on the retina may be the first time the light rays projected on the retina are collectively referred to as a picture. This reference to the projection as a picture is reinforced by Kepler's comparing the projections from the rays onto the retinal surface to small brushes as a painter might use (Costall, personal communication, January 15, 2021).

Arguably, Kepler's important discovery is not a break with the optics of the medieval period but instead one that works within that tradition. Lindberg writes that “Kepler labored within a conceptual framework given him by the medieval perspectivists...” (p. 207). In that regard the two-dimensional retinal image is *conceptually* continuous with the point-by-point cross-section of light posited to be projected at the anterior glacial surface of the lens, thereby corresponding to the two-dimensional base of the visual cone at an object's surface. And that view, as we saw, is based on the assumption that held since antiquity that visual perception is based on the reception of images (see, Smith, 2015). To the extent that contemporary theories of vision adopt the stance that what is received by the eye are images, that claim has a long history indeed.

3.1 | The invention of artificial perspective

Before tracing this idea of image-based perceiving further over time, it is necessary to introduce into our account at this point developments in the visual arts occurring *nearly two centuries prior* to Kepler's discovery. I refer here to the invention of artificial linear perspective, about which there is an extensive literature (e.g., Edgerton, 2009; Gombrich, 1960; Hyman, 2006, Ch. 10; Panofsky, 1991).

It is critical to distinguish *artificial perspective* from *natural perspective*, which are frequently conflated in the literature of modern perceptual psychology. Beginning with the latter, the application of Euclidean geometry for the purposes of describing the path of light rays between the individual's eye and the object perceived is properly referred to as *natural perspective*. It is attributable to the layout of the environment and the objects in the field of view that can be perceived from a particular point of observation (see Figure 1). Among the most commonly cited examples of it today are the parallel rectilinear edges of train tracks appearing to narrow as they recede into the distance. In this case, there is a gradual reduction in the visual angle projected from any two adjacent points on the parallel edges of the path to the eye of a stationary perceiver.

Because this visual pattern is a to features of the environment taken in relation to the position of the perceiver, art historian Edgerton (2009) rightly states, “there is nothing in original *perspectiva* theory [i.e., *natural perspective*] that had anything to do with the visual arts” (p. 22). That is, we are not referring here to a technique for transforming the appearance of the environment onto a flat surface, but rather offering a geometric analysis of the environmental layout along lines of sight. Leonardo, for one, wrote about this difference (see, Smith, 2015, p. 312).

With that in mind, natural perspective is to be distinguished from the *technique* that was developed and put into wide practice beginning in the 14th century to create the appearance of depth on a flat surface—a technique properly called *artificial perspective*.² Historians are in agreement that artificial perspective techniques were developed by Brunelleschi (1377–1446) for artistic purposes to display or represent depth on a flat surface. Though natural perspective is a result of the application of a human invention, namely, geometry, to the perceived layout of the environment, artificial perspective is wholly a human invention from start to finish.

From available accounts, Brunelleschi's initial effort to employ artificial perspective was a small painting on wood of a building in Florence, The Baptistery. He utilized orthographic projection which is commonly employed by architects in creating plans that offer separate two-dimensional views of a built structure from the front, top, and

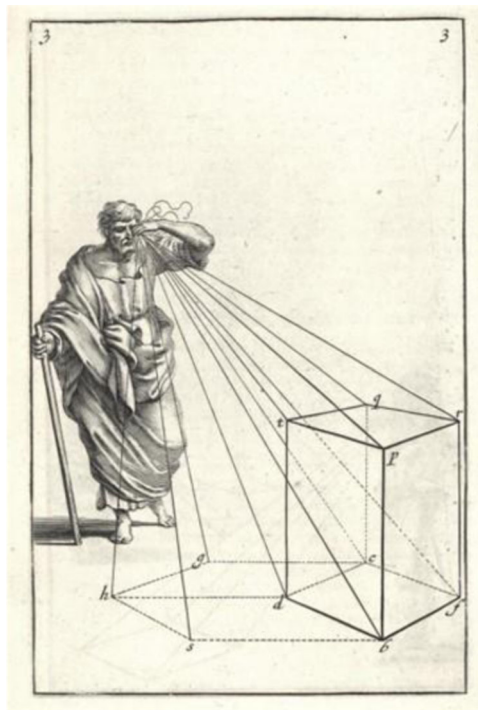


FIGURE 1 Natural perspective. From Bosse, A. *Manière universelle de Mr Desargues* (1648) <https://www.pinterest.pt/pin/744290275886828480/>

side. It is likely to have been more than a coincidence that Brunelleschi had been trained as an architect, with his most famous construction being The Duomo in Florence (King, 2000). Although the precise procedure that he used in developing linear perspective continues to be debated among scholars, Panofsky (1991) and Edgerton (2009) among others have proposed that Brunelleschi translated by way of orthographic projection the appearance of a three-dimensional object from both a plan and elevation view onto a single two-dimensional surface (also see Sedgwick, 1980; Smith, 2015). The point on the picture plane at which the set of projection lines converges is the vanishing point.

On the heels of Brunelleschi's demonstrations of perspective displays, Alberti (1404–1472) proposed in his book *De pictura* (1452) another technique for rendering perspective drawings. A frame (imagined or real) can be inserted as a cross-section of the visual cone extending from the object to be rendered to the artist's point of vision (see Figure 2). Alberti proposed that that the frame should be taken as “an open window through which the subject to be painted is seen” from the standpoint of stationary viewer (quoted in Friedberg, 2006, p. 27). The Latin roots of the term perspective mean “seeing through” in the sense of looking at the world through the frame of a window from a stationary viewing position (Panofsky, 1991). So-called “Alberti's window,” then, is a cross-section of the visual cone from the field of view to the perceiver's eye.

To facilitate the translation of the form of the object onto the cross-section that is the frame, Alberti suggested that its surface be covered with a meshwork of horizontal and vertical perpendicular threads, a *velo* (a veil), which would regularly section the surface into a grid (Edgerton, 2009; Friedberg, 2006; Smith, 2015, p. 304). The use of a *velo* is illustrated in several woodcuts by Dürer from the 1580s (see Figure 3). It is noteworthy that some of these woodcuts also show the artist making use of a vertical reference rod for fixing his line of sight, establishing a *stationary viewpoint* when drawing (Hyman, 2006). With reference to a slightly different framing technique,

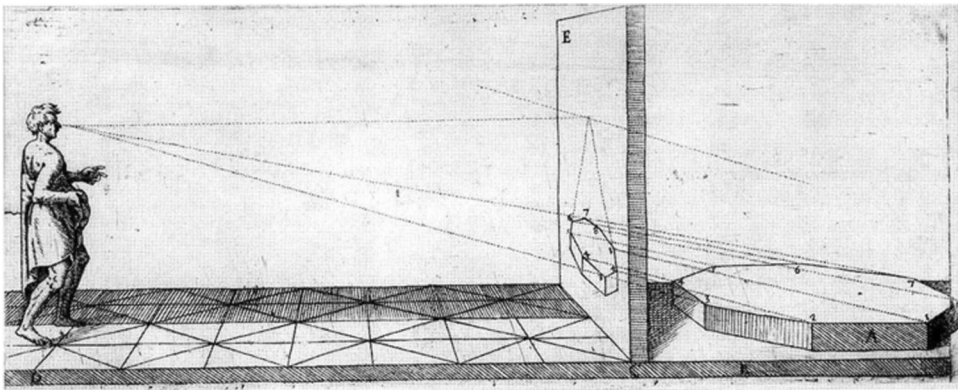


FIGURE 2 Artificial linear perspective. From Jacopo Barozzi da Vignola—Perspective diagram (1583) https://commons.wikimedia.org/wiki/File:Jacopo_Barozzi_da_Vignola_-_Perspective_diagram_-_WGA25090.jpg

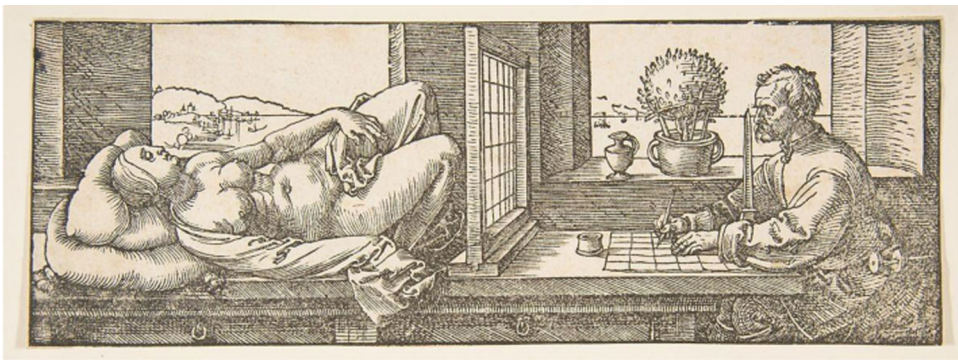


FIGURE 3 Albrecht Dürer—Draftsman making a perspective drawing of a reclining woman (ca. 1600) <https://www.metmuseum.org/art/collection/search/366555>

Leonardo indicates that “the head is fixed by an instrument in such a way that the head may not be moved at all” (quoted in Friedberg, 2006, p. 39).²

Though Alberti's window was indeed an important innovation for the creation of drawing and paintings in perspective, it should be recognized as an extension of prior conceptualizations. The idea of the visual cone or pyramid positioned such that the object seen is framed at the cone's two-dimensional base is repeated with Alberti's window introduced by the artist as an “artificial” two-dimensional cross-section of the cone.

Paintings and murals that employed linear perspective techniques in grand style, such as Raphael's *The School of Athens* (1509–1511) and Leonardo's *The Last Supper* (1495–1498), are among the most well-known works of the era (see Figure 4). For many they represent the high point of Renaissance painting.

Considered more broadly, the invention of artificial linear perspective is a symptomatic of a transformative era in Western history—a transition from a Medieval cosmic hierarchy with human concerns located at a low rung of significance (Cassirer, 1963) toward an ethos where the individual person's experiences, thoughts, and motives moved to a focal place in understanding human affairs. Individuals' efforts based on sensible experience and logical reasoning slowly came to be embraced as the means for arriving at structures of universal order and truth in the world. This momentous era of cultural change in the West has commonly been referred to as *Humanism*. The cultural critic Berger (1972) describes the relationship between linear perspective and Humanism:



FIGURE 4 Raphael—School of Athens (1509–1511) https://en.wikipedia.org/wiki/The_School_of_Athens

The convention of perspective, which is unique to European art and which was first established in the early Renaissance, centres everything on the eye of the beholder.... Perspective makes the single eye the centre of the visible world.... The visible world is arranged for the spectator as the universe was once thought to be arranged for God.... it structured all images of reality to address a single spectator, who unlike God, could only be in one place at a time. (p. 16)

3.2 | Conflating natural perspective and artificial perspective

The preceding brief discussion of the difference between natural and artificial perspective is offered in an attempt to clarify a matter of considerable confusion in the contemporary visual perception literature, where we commonly find “linear perspective: listed as a monocular cue on the retina for depth *in the environment*. It is important to recognize, however, that such assertions typically conflate “natural perspective” and “artificial perspective” and in doing so, obscure the differences between them. J. J. Gibson (1979) explains:

When Renaissance painters discovered the procedures for perspective representation, they very properly called the method *artificial perspective*. They understood that this had to be distinguished from natural perspective that governed ordinary perception of the environment. Since that time we have become so picture minded, so dominated by pictorial thinking, that we have ceased to make the distinction. (p. 70)

But what is the problem with conflating the two? After all, aren't the base of the visual cone from a natural perspective orientation and the cross-section of the visual cone that is Alberti's window isomorphically equivalent from the standpoint of projective geometry? That may be accurate when the perceiver remains in a fixed, stationary position. But when an individual is viewing the layout of the landscape (natural perspective) rather than a picture, she can change the direction of view by turning in place (rotational movement) or changing the viewing position (translational movement). With artificial perspective, the frame of Alberti's window constrains the field of view and the world that is visible to the perceiver is frozen.

But to confuse pictorial perspective with natural perspective is to misconceive the problem of visual perception *at the outset*. The so-called cues for depth in a picture are not at all the same as the information of a surface layout [in ordinary perception]... although pictorial thinking about perception tempts us to assume that they are. Pictures are artificial displays of information *frozen in time*.... (J. J. Gibson, 1979, pp. 70–71; emphases added)

Importantly, changes in viewing position produce changes in the field of view that reveal features that are *unavailable from a stationary viewpoint*. To illustrate using contemporary terminology, transformations in what is available to be perceived in the context of movement reveal, for example, invariant structures, motion parallax, dynamic occlusion, and optic flow—none of which are present in a “frozen” cross-section of the light provided through artificial perspective (Heft, 2020). We will return to this issue at a later point.

Here, then, the origins of the two “enduring assumptions” discussed earlier—taking the two-dimensional array as the beginning point of an account of visual perception, and adopting the stationary perceiver as the normative case—come clearly into view, as do some of their limitations. Clark (2007) insightfully points out “Renaissance attempts to make linear perspective the pictorial equivalent to vision... [eventually led] to the view that vision *itself* was pictorial” (p. 16). But in doing so, as Gibson noted, a wide range of sources of information in a changing visual field is entirely overlooked. Not only does a pictorial account constrain our considerations about the available stimulus information for vision, but critically, it separates seeing from acting, two processes that J. J. Gibson (1966) proposes to be functionally synergistic. We will return to this point below as well.

3.3 | The perspectivist fracturing and freezing of time

The realistic effects on a flat surface that linear perspective makes possible come at a cost. Perhaps with works in mind such as Assyrian reliefs (880–612 BCE) where a sequence of events is represented, and instances of a series of Roman murals painted adjacently (circa 1st century³), the contemporary painter/photographer David Hockney has commented in an interview:

Before the fifteenth century... you have [in pictorial representation] lateral movement: the story is told moving about. You move along, and that way things appear less solid... In doing that you have the advantage of time. But in wanting to make it solid [through perspective drawing] you must stop time, that's the problem, the core of the problem, isn't it? In creating depth you.... ['abolish time', interjected the interviewer]. (Joyce, 1988, pp. 37–39)

Movement and time have been set aside in the interest of what is in effect an *illusion* of depth. It is “*a fracturing of times with a single space*” (Friedberg, 2006, p. 36, emphasis added).

And yet “abolishing time” is not a necessary consequence of attempting to portray events on a two-dimensional surface as evidenced by the examples cited above.⁴ Even more striking efforts to maintain temporality can be found in Asia with its long history of Chinese and Japanese scroll paintings.

A notable example is a Chinese landscape painting (datable to 1698), one panel of which is currently on exhibit at the Metropolitan Museum of New York. (see Figure 5). The panel is 45 feet long when opened and is but the third of a total of 12 panels depicting an emperor's tour of a region of the country, which when placed end-to-end extend for 700 feet. These scrolls were created to be experienced by a viewer over time while moving along their lateral extent. In contrast, Western perspectivist paintings are intended to be taken in from one observation point, peeling the temporal dimension away in the interest of the illusion of spatial depth.



FIGURE 5 Wang Hui—The Kangxi Emperor's Southern Inspection Tour, Scroll Three: Ji'nan to Mount Tai MET DP105293.jpg

4 | THE RETINAL IMAGE AND ITS LEGACY OF THE SENSORY CORE

The consequences of applying artificial linear perspective to theories of everyday visual perception, along with its underlying assumption that perceiving is based on the capturing of images on the retina, have been enormous and lasting. If it is assumed that visual perception begins with a two-dimensional image on the retina, it is obvious that a full account of visual perception must go well beyond that, as even Kepler recognized with his discovery of it in the early 17th century. The standard approach to addressing this gap between the visual properties conveyed by the retinal image and those of visual experience is to assume that mental processes beyond the retinal image account for our everyday experience. Hatfield & Epstein (1979) propose that *what such mental processes work on* is an hypothesized “mental correlate of the retinal image” (p. 363) which they refer to as “the sensory core.” They describe the sensory core as providing “the raw material from which the mind *generates* the three-dimensional visual world” (p. 363, emphasis added). If perceiving entails operating on a mental correlate of the retinal image, then the possibility that perception of the environment is unmediated and direct is ruled out as a matter of course.

In his extensive survey of post-Renaissance theories of spatial perception, Hatfield (1990) reiterates the central role that something like the sensory core has played: “Even though phenomenal access to the sensory given seemed precluded, there was nearly universal agreement regarding its characteristics. The predominant view was that it is a mental representation of the retinal image” (p. 35). In other words, although by definition perceivers are unaware of the existence of the sensory core, *its place in perceptual processes is taken for granted owing to the assumption of two-dimensional retinal image*. With that as the starting point for visual perception processes, surely among the most vexing problems for perceptual theory, as we saw above, is how to account for the perception of distance and, in turn, phenomena that depend on it, such as object size and object shape.

Berkeley's (1709) *An Essay Toward a New Theory of Vision* has been especially influential in establishing the two-dimensional array as the starting point for a theory of visual perception.⁵ In an oft-quoted passage, he wrote: “It is, I think, agreed by all that distance, of itself and immediately, cannot be seen. For distance being a line directed end-wise to the eye, it projects only one point in the fund of the eye” (p. 13). On those grounds, Berkeley claims that vision alone cannot account for the perception of distance. He famously proposed that the experience of spatial extent is generated from the association of sensations projected on the retina with sensations from motor responses, such as pupillary accommodation and binocular convergence.

But what is the basis for Berkeley's initial claim "that distance, of itself and immediately, cannot be seen"? It cannot be an empirical claim: What is projected on the "live" retina cannot be seen either by an outside observer⁶ or logically by the individual himself. Apparently, it was just an accepted belief at the time of his writing:

This view... had been widely accepted before Berkeley wrote, and it was repeated as a matter of course by a variety of authors throughout the eighteenth century.... When Berkeley built his theory of vision on the view that the proper object of sight is a picture-like correlate of the bi-dimensional retinal image, he offered no justification for his position beyond an appeal to previous visual theory. (Hatfield, 1990, p. 35)

For the most part, it continues to be "repeated as a matter of course" to the present day.

By the early and mid-19th century, discussions of the basis for space perception shifted emphasis toward physiological considerations; and owing to the legacy conveyed by Berkeley, the character of sensations produced by retinal stimulation were assumed to be "depthless." Among the most influential contributors to this physiological turn during the first half of the 19th century was Johannes Müller (1801–1858), later followed by his student Hermann von Helmholtz (1821–1894).⁷

Helmholtz's *Handbook of Physiological Optics* (published over the period 1924–1925) was to have a long-term impact on later 20th century theories in visual perception. We find in his treatment of "space perception" continuation of the assumption that the visual perception from the outset is lacking in three-dimensional qualities. Consistent with Lotze and other prominent 19th century theorists, Helmholtz claimed that visual experience of a three-dimensional world arises out of the combination of sensations, each of which is originally nonspatial or "depthless" while varying only in quality (e.g., color) and intensity. In most accounts, sensations acquire the spatial character of directionality as a "local sign" only as a result of their association with motor sensations, such as those that accompany efforts to fixate the direction of a light source on the retina as well as movements of the hand in coordination with eye movements.⁸ Collectively, these motor sensations give rise to a mental representation with spatial qualities; and subsequently, received visual sensations can be employed to "unconsciously" infer spatial volume based on such a mental representation from originally nonspatial sensations. Although Helmholtz identified his commitments as being in the Kantian tradition, which is born out in his well-known work on color experience and auditory sensations, the empiricist influence of Berkeley on Helmholtz's account of visual perception is unmistakable. It would be difficult to overstate the influence of Helmholtz's writings about visual space perception on subsequent generations of perceptual theorists (e.g., Rock, 1985).

Helmholtz's views on spatial perception which adopt the position that visual perception begins with a two-dimensional display on the retina became the mainstream view for most English-speaking psychologists over the first half of the 20th century. A particularly explicit expression of this viewpoint can be found in (Woodworth, 1921) *Psychology*:

The retina is a screen, and the stimulus that it gets from the world outside is like a picture cast upon a screen.... A single, motionless eye receives a picture similar to one painted on a canvas... (Woodworth, 1921, pp. 398–399)

Among the most influential works conveying Helmholtz's ideas on perception to later generations were Boring's landmark books *The History of Experimental Psychology* (1929) and *Sensation and Perception in the History of American Psychology* (1942). Especially important for our purposes here, Boring observed nearly at mid-century in the latter book that whereas Helmholtz had given primary significance to physiological mechanisms such as accommodation and convergence, and only secondary weight to pictorial cues for depth, "[i]n the present century the pendulum has started to swing back; the classical 'secondary criteria' [i.e., pictorial cues] are now being regarded as the primary dynamic mechanisms in visual perception..." (p. 263).

This observation is noteworthy because it anticipates a great deal of work in the 50's and 60's, which paved the way for much of contemporary "cognitive science," where we find that the Keplerian perspectivist assumption that vision is in the first instance primarily pictorial (i.e., an image) takes center stage. To cite just one influential example, Marr (1982) proposed in his widely-read book *Vision* that visual perception proceeds from a two-dimensional array of light on the retina, from which the fundamental components of the projected image, for example, edges, curves, are extracted, resulting in a "primal sketch." The latter is then conveyed to a so-called $2\frac{1}{2}$ D sketch where the orientation, texture, and depth information is produced, with that information then passed onto a level where a three-dimensional model is generated. Reflecting on Marr's proposal two decades after it was introduced, Glennerster (2007) wrote that the conception that vision involves deriving properties of the perceived world initially from *images* of it is "at least a widely accepted mantra in biological vision research" (p. R397).

How far have we come from Kepler's initial claim that an image is projected on the retina, and that perceiving proceeds based on the reception of that image? Although the particulars of the "upstream" vision processes have been much elaborated over centuries, Kepler's initial assertion has endured. We see this in a recent overview of the field by a leading researcher of sensory processes: "The combined result of refraction at the cornea and through the crystalline lens produced an image of each point *in the scene* as a corresponding point on the retina—that is a retinal image of the scene is created" (Cornsweet, 2017, p. 91, emphasis added).⁹ Without question, much has been learned about neurophysiology and neurochemistry of sensory processes, not to mention the visual tract and visual processes in the brain. But the most basic assumptions about the starting point for *visual perception* itself have largely remained unchanged since the Renaissance.¹⁰

Some, however, have judged this enterprise misconceived from the beginning. Wartofsky (1973) claims that by taking geometric optics as the foundation for an account of visual perception, theorists mistakenly took "a theory of image-formation for a theory of perception" (p. 197). To begin in this way is to allow abstractions such as light rays and the geometric optics that describe their paths— useful as they may be—to drive inquiry rather than the psychological phenomena of visual perception themselves. The judgment is consistent with the opinion Berkeley, no less, offered around 250 years ago:

To explain how the mind or soul of man simply sees, is one thing, and belongs to Philosophy [and now psychology]. To consider particles as moving in certain lines, rays of light refracted, or reflected, or crossing, or including angles is quite another thing, and appertaineth to Geometry. To account for the sense of vision by the mechanism of the eye is a third thing, which appertaineth to Anatomy and experiments. These two latter speculations are of use in practice, to assist the defects and remedy the distempers of sight, agreeably to the natural laws contained in this mundane system. *But the former theory [i.e., "to explain how the mind simply sees"] is that which makes us understand the true Theory of Vision.* (Berkeley, 1733, p. 76; emphasis added)¹¹

In the remainder of the paper, I will trace an alternative line of inquiry about visual perception that emphasizes *what we see*. This will extend from William James' psychological writings about perception that try to reorient theory toward what we experience; to visual artists, such as Cezanne, Picasso, and Braque, who challenge viewers to reconsider the very character of visual experience, especially in the wake of new 19th century technologies; to James Gibson's ecological approach that rejects the claim that visual perception is based on the reception of pictures or images, and instead takes as its starting point the on-going, dynamic nature of perceptual processes in relation to environmental structures and features.

5 | WILLIAM JAMES' CHALLENGE TO BERKELEY-HELMHOLTZ ACCOUNT OF VISUAL PERCEPTION

The standard or received approach to visual perception proposes a linear sequence from (a) sensory (retinal) stimulation producing (b) sensations which then are utilized as the "materials" for (c) intervening processes that give rise to perceptual experience. In broad outline, it has been the template for most theories of visual perception since at least Helmholtz's day, and as we have seen, it is a legacy of perspectivism.

In the landmark work *The Principles of Psychology*, James (1890/1981) challenged the claim that sensations are either temporally before percepts, or even constituents of them (see Chapter XVII, in particular). His argument is best appreciated in light of his early training as a painter, which preceded his formal studies in medicine and his decades of work on psychological and philosophical topics (Bjork, 1983; Richardson, 2006). It is likely that those experiences gave him a decidedly more phenomenological orientation toward visual experience than most of his contemporaries in experimental psychology who came to perception from a background in sensory physiology. As a painter in training, James needed to attend carefully to *the appearance of things*. It is also worth bearing in mind that James is writing near the end of the 19th century with many of the transformative intellectual and technological changes of that century as a background. We will consider those circumstances shortly.

A distinctive attribute of James' writings throughout his career was his inclination to take the character of immediate experience as the starting point for inquiry (Heft, 2001, 2003). We can even see this in the opening sentence of *The Principles*: "Psychology is the Science of Mental Life, both of its phenomena and of their conditions" (James, 1890/1981, p. 15). It is with the *phenomena* of mental life that psychological investigations should begin.

With that in mind, James considers sensations and percepts as they appear in *immediate experience*. He writes:

The nearer the object cognized comes to being a simple quality like "hot," "cold," "red," "noise," "pain," apprehended irrelatively to other things, the more the state of mind approaches pure sensation. The fuller of relations the object is... the more unreservedly do we call the state of mind a perception..., *Sensation, then, so long as we take the analytic point of view, differs from Perception only in the extreme simplicity of its object or content.* (James, 1981, pp. 651–652, emphasis in original)

Sensations are observable qualities in immediate experience, not unobservable "sensory core" materials. Most critically, they cannot comprise the material from which *spatial* experience is constructed, even as local signs as Helmholtz proposed, because they are originally nonspatial in character. James asks how an account that begins with nonspatial sensations can "explain the *extensive quality itself*, account for it, and evolve it, by the mere association together of feelings which originally possessed it not." (p. 901, emphasis in original). Simply stated, "How, it may be asked, can association produce a space quality not in the things associated? How can we by induction or analogy infer what we do not already generically know?" (pp. 908–909). He views this problem as a critical shortcoming for the Berkeley–Helmholtz approach; and it can be extended to all theories that claim spatial qualities are added by post-retinal processes to immediate experience lacking in spatial qualities.¹²

5.1 | The *psychological immediacy* of "space"

Instead, James offers the phenomenological claim that "[t]he third dimension forms an *original element* of all our space-sensations" (p. 900, emphasis added). He reports that "all objects of sensation are [originally] voluminous in three dimensions" (p. 847); that "[t]he field of view is always a *volume-unit*" (p. 848); and that "[s]ensations have "the *original sensation of space*, out of which all the exact knowledge about space that we afterwards come to have is woven...." (James, 1981, p. 777).

We might consider these observations from the vantage point of James' early years as an artist in training, when he was challenged to transform the immediate appearance of a voluminous space onto a two-dimensional canvas. Looking back, this is precisely the problem that Brunelleschi attempted to solve when he invented artificial linear perspective. Later, in an historical reversal of antecedent circumstances, Brunelleschi's solution to the problem of representing the experience of voluminous space on a two-dimensional surface—namely, the technique of artificial perspective—is taken centuries later by some perceptual theorists to be an antecedent of the experience of spatial extent itself, being a visual “cue” for inferring three-dimensional space from a two-dimensional retinal image.

Still, it is one thing for James to claim that immediate experience has a voluminous quality, but short of appealing to a Kantian *a priori*, which he is loath to do,¹³ how is it possible to ground this claim *in visual experience itself*? Here, James takes an important conceptual step by asking the reader to consider “a more complicated case” than the static projection of an object, but instead an object in *motion*. He offers the example of a stick rotating around a horizontal axis in the “distal plane” of the field of view, and describes the resulting perceptual phenomenon event as follows:

In this movement, the stick's image will grow progressively shorter; its farther end will appear less and less separated laterally from its fixed near end; soon it will be screened by the latter, and then reappears on the opposite side, and finally on that side resume its original length. Suppose this movement to become a familiar experience; the mind will presumably react upon it... and consider it the *movement of a constant object* rather than the transformation of a fluctuating one.... Thus we find that given amounts of visual depth-feeling become signs of fixed amounts of the visual breadth-feeling. The measurement of distance is, as Berkeley truly said, a result of suggestion and experience. But visual experience alone is adequate to produce it, and this he erroneously denied. (p. 849)¹⁴

This passage is remarkable for many reasons, but perhaps most of all because James is *abandoning the pictorial claim that perceiving begins with the projection of an image or a static picture-like display to the eye*. This is an imaginative leap reconfiguring what has long been taken to be the beginning point for an account of visual perception. With this step, he invites us to consider what visual experience in its immediacy *over time* is like. In the present example, he finds that the experience of spatial “volume” accompanies or is a concomitant of the visual transformation (e.g., the rotation) of the object in immediate experience. In this respect, “space” is directly experienced—and it is so originally, at least, over time—rather than being mediated by postretinal operations on the sensory core.

This supposition would logically extend to reciprocal instances when an object is fixed in place and the perceiver is moving in relation to it. And note that instances of a moving perceiver would run counter to a pictorial approach to visual perception as well. Recall that the artist was instructed to adopt stationary position when fashioning a perspective drawing (Figure 3); and likewise, in order for a viewer to maximize the experiential effects of artificial linear perspective, ideally, he is to assume a stationary observation position with respect to the two-dimensional surface of the canvas.¹⁵ In contrast, for James an observation point is at best a temporary position taken by an active perceiver. He writes: “The world experienced... comes at all times with our body at its centre, centre of vision, *centre of action*, centre of interest.... The body is the storm centre, the origin of coordinates, the constant place of stress in all that experience-train” (James, 1976, footnote, p. 86, emphasis added). Movements of the body are experienced in relation to space as a “volume-unit” because “space” is a concomitant of experiencing the body moving (Sheets-Johnstone, 2016) or in relation to features of the environment (J. J. Gibson, 1979).¹⁶

6 | CREATING VISUAL REALISM THROUGH ARTIFICE

The *invention* of linear perspective in the 14th century makes it clear that the production of realistic-looking drawings and paintings does not involve a mere copying of forms onto a projection surface. *The appearance of realism is created rather than copied* (see, Snyder, 1980). And yet when the retinal image is conceptualized as a projection of light on a surface that must be “interpreted” by processes in the visual stream, that projection tends to be treated as if the layout of the environment is mostly transferred in an isomorphic manner to the retinal surface. That simple isomorphic assumption belies the history of perspective representation which took considerable ingenuity to accomplish.

It is noteworthy, however, that even as perspectivism continued to influence thinking about visual perception in philosophy and later psychology to the present day, the techniques of linear perspective did not continue to play a dominant role in the visual arts, even when the explicit goal remained the creation of realistic-looking paintings. Attention shifted markedly in the 17th century, particularly among the Dutch realists, to detailed studies of the interaction of light with surfaces and volumes, and later the interplay of space and time. The art historian LaPorte (1949) observed: “By the first half of the seventeenth century a consistent interpretation of nature is completed *which is no any longer based upon the application of mathematical perspective* but on varying intensities of light” (p. 247, emphasis added). In other words, even while perspectivism and its assumptions continued to influence thinking about visual perception, particularly in 19th and 20th century psychology, reliance on perspectivism among visual artists had long been on the wane. Developments in 19th century science and technology prompted even further changes in the visual arts.

6.1 | Science, technology, and perceiving “time”

Broadly stated, the 19th century was a period when developments within the sciences led to an increased focus on *change* and *process* inherent in the dynamic character of the natural world. Lyell's writings on geological changes in earth history (uniformitarianism) followed by Darwin's revolutionary work on natural selection were among the more notable contributions to this overall shift toward a conception of a dynamic natural world. Near the end of the century William James' emphasis in *The Principles* on temporality, as is most obvious in his treatment of mental experience ('the stream of thought'), resonates with this shift in thought.

A heightened focus on change and time was also driven by new technologies, the photographic camera and railways, in particular. Photography got underway in the 1820s and 30s. The photographic camera is, in effect, a camera obscura which has a rear interior surface that is altered when exposed to light, and with proper post-exposure treatment the image projected onto that surface can be ‘fixed’ (see Gernsheim, 1986).¹⁷ By the 1870s photographers such as Marey and Muybridge created a sensation by photographically capturing successive, discrete moments of on-going events, such as animal and human movement. When those photographs were displayed adjacently as a succession of “stills,” *the passage of time* was tangibly made visible.

Prompted at least in part by these experiments in capturing motion, visual artists began to portray views of the environment and of individuals under fleeting circumstances. Well known among many efforts of this kind were French Impressionist studies of how shifting conditions in lighting at different times of day, in different seasons, and under different viewing circumstances affect the momentary appearance of landscapes and buildings. Some of Monet's most well-known Impressionist paintings (e.g., the Haystacks and Rouen Cathedral series) are of this nature.

There are also efforts to represent scenes or individuals at “instants in time” and often at atypical moments in action, under atypical lighting conditions, or from atypical and notably temporary viewing positions. Some of Degas'

paintings are of this nature; and surely it is not coincidental that he was an avid photographer (Daniel, 1998). Collectively, these kinds of works were a break from paintings that offer a universally representative or a canonical moment, and instead are intended to capture a particular, nearly idiosyncratic instant in time—a moment (that like an instant in James' stream of thought) would not recur exactly in the same manner again. In all of these efforts in visual depiction, time has become an essential dimension of a two-dimensional work. It is still “frozen time,” but *the moment in time* and its fleeting character are foregrounded.

A transformation in cultural awareness of the passage of time has also been attributed to the experience of traveling by train, which like photography began in the 1820s (Schivelbusch, 1986). Moreover, moving through a landscape at faster speeds than ever before not only heightened the experience of time, but by reducing the time needed to traverse particular distances, it also experientially resulted in a shrinking of space. In addition, the necessity for coordinating train schedules forced a synchronization of clocks at different locations (Galison, 2003). As a result, space and time ceased to be fixed orthogonal dimensions, but instead become somewhat pliable, relative, and intertwined. This shift in awareness is formalized in relativity theory at the outset of the 20th century (Hoffmann, 1983; Vargish & Mook, 1991).

It is decidedly not the case that those works in the visual arts attempting to integrate time into a pictorial representation either by accentuating particular viewing instants or by catching events on the fly are unrealistic-looking as compared to classic “timeless” perspective painting. Arguably they are more realistic in the sense that they are not attempting to portray an idealized view of how things should look relative to some convention (Costall, 1993). Rather they are intended to capture a segment of the world at a moment in time *as it may be experienced by a particular individual under specific circumstances*. Reality is being portrayed in its multitude of appearances, each equally valid. A pluralistic point of view is to become an essential feature of James' pragmatism and his ontology (James, 1920, 1996).

6.2 | Abandoning the pictorial picture plane as a field of view for “the realism of the psychological process itself”

It is in relation to this background of attempts to represent the perceived world at a moment in time that the artist Cezanne took what is widely acknowledged as a revolutionary step to create works that displayed a landscape or a still life *from more than one observation point, hence at different times, within a single picture frame*. These experiments were to be transformative because they “destroyed forever in art the possibility of a static view of nature” (LaPorte, 1949, p. 59).

To illustrate the inclusion of multiple viewpoints in a single picture frame, consider Cezanne's “Basket of Apples” (1893) in which different portions of the painting appear as if they were viewed from different observation points (see Figure 6). Linear discontinuity attributable to different observation points is perhaps most striking with respect to the front edge of the table, where the visible edges seem misaligned while being hidden by a cloth.

LaPorte (1949) describes works such as these by Cezanne as “the final point of departure from occidental perspective as first developed in the fifteenth century...” (p. 245). Fry (1966) explains “Cezanne has broken with the Renaissance tradition [of]... illusionistic stage space of one-point perspective,” and in its place he has adopted “the realism of the psychological process itself” (p. 14, emphasis added). After centuries, the perceiver has been dislodged from a single, and idealized, stationary viewpoint that had been assigned to him and is recognized as experiencing the world from a moving point of observation in the course of everyday life. Moreover, the “Basket of Apples” portrays a perfectly mundane setting, as is the case also with Cezanne's landscape series, not a stylized nor idyllic one. This is an embrace of the realism of ordinary experience.



FIGURE 6 The Basket of Apples—Paul Cezanne (1895)—The Art Institute of Chicago: Art Access

6.3 | From time-bound representations to perceiving over time

Even more radical experiments in depicting the changing relationship between a landscape or a display of objects and a perceiver emerge with the early 20th century movement of Cubism, initiated jointly by Picasso and Braque. The self-consciously dynamic character of their goals has been described by their contemporary Apollinaire: “They no longer painted an object from one perspective, but rather layered views from many angles to capture the subject from all sides” (Apollinaire & Emert, 2010, p. 29). The Cubist painter Metzinger (1910) describes this effort as having “uprooted the prejudice that commanded the painter to remain motionless in front of the object, at a fixed distance from it, and to catch on the canvas no more than a retinal photograph... They have allowed themselves to move round the object, to give... *a concrete representation* of it made up of several successive aspects” (p. 66, emphasis added). Through this approach, they hoped to convey a fuller experience of space or layout than possible from a fixed observation position.

Braque was clear on his intent: “The whole Renaissance tradition is antipathetic to me. The hard-and-fast rules of perspective which it succeeded in imposing on art was a ghastly mistake which it has taken four centuries to redress... it is simply a trick—a bad trick—which makes it impossible for an artist to convey a full experience of space” (Miller, 2001, p. 130, quotation from Richardson, 1991).

We can see an early expression of the Cubist juxtaposition of multiple views of a figure from different vantage points—views that would normally occur at different times when moving with respect to it—by considering the two central figures in Picasso's ground-breaking *Les Femmes d'Alger* (see Figure 7). The frontal view of their faces includes profiles of the nose; and more drastically with the figure on the lower right, a frontal view of the face/mask is combined with a view of the figure's back no less. Other and more experimental forms of Cubism that were to follow were less pictorial and more structural and analytical, presenting subject matter as multiple planes or facets viewed from multiple angles (see Figure 8).

In relation to trends in the sciences, these works were understood at the time of their development as “the artist's response to the changed preconditions of science regarding space and time” (Apollinaire & Emert, 2010, p. 29). This parallel is not quite “as surprising as it may seem if one considers that *both physics and painting are concerned with the same problem, namely the nature of space*” (LaPorte, 1949, p. 244, emphasis added). Physics had been changing since at least Faraday's 1820s work on electromagnetic fields and the later the emergence of field theoretical thinking. Those advances led eventually to abandoning the view that space and time are fixed and separate dimensions, but instead are functionally



FIGURE 7 Pablo Picasso—*Les Femmes d'Alger* (1935) <https://www.moma.org/collection/works/79766>

related and variable (Hoffmann, 1983; Mook & Vargish, 1987). For example, spatial properties, such as the length of an object, varied as a function of the speed of the measuring observer. An epistemological implication of this discovery is that there is no single privileged location or one privileged set of observation conditions from which to measure or otherwise describe a fixed reality as perspectivism implies. Rather there are multiple legitimate locations and conditions for taking stock of things. In this respect, reality is not one all-encompassing unitary entity, but a manifold of multiple possibilities *taken relative to an observer*.

Considered in relation to everyday realities, it can be argued that Cubism and related movements (i.e., Futurism) rather than distorting reality—as Riviere (1912) remarked, “[artificial] perspective is as accidental as lighting” (quoted in Costall, 1993, p. 340)—are bringing the visual arts *closer* to everyday experience. The works of the Cubists “have as great a claim to realism as do the tricks of the Perspectivists” (Hagen, 1980, p. 11); and “[p]robably only a further shift of the artistic reality level is needed to make the Picassos, Braque, or Klees *look exactly like they things they represent*” (Arheim, 1954, quoted in Hagen, 1980, p. 11). In their early book on Cubism, Gleizes and Metzinger (1912) remark that as the notion of perceiving over time from a moving point of observation becomes more commonplace “[t]hen the fact of moving around an object to seize several successive appearances... will no longer make thoughtful people indignant” (as quoted in LaPorte, 1949, p. 246).

7 | IMPLICATIONS FOR A THEORY OF PERCEPTION: PERCEIVING AND ACTING

[F]or the first time since the Renaissance, a new conception of space leads to a self-conscious enlargement of our ways of perceiving space.... Cubism breaks with Renaissance perspective. It views objects relatively that is from several points of view, no one of which has exclusive authority,



FIGURE 8 Pablo Picasso—Portrait of Daniel-Henry Kahnweiler (1910). Art Institute of Chicago

and in so dissecting objects it sees them simultaneously from all sides—from above and below, from inside and outside. (Gideon, 1944 as cited in Freidberg, 2006, p. 118)¹⁸

What lessons might be learned from this “new conception of space” for theories of visual perception? For one thing, it places front and center the idea of the perceiver as an active agent exploring the surrounding environment layout. That lesson turns out to accord with recognition within in the life sciences early in the 20th century that self-directed action is perhaps the defining property of animate life (Jennings, 1906). That is, action rather than vision is the essential function of organisms, and that quite likely vision in organisms evolved in the service of self-directed movement. On these grounds, the dynamic character of vision, that is, perceiving over time, would be the norm, while perceiving from a fixed observation point (apart from adaptively “freezing” to avoid being seen) is comparatively less common.

The perceptual psychologist James Gibson began to consider perceiving from a moving point of observation as early as the 1930s, and like some Gestalt psychologists (e.g., Lewin) framed it in terms of a field-theoretical perspective (J. J. Gibson & Crooks, 1938). Not long after that, during the war, he studied perceiving the ground surface while moving from the point of view of a pilot landing a plane (e.g., J. J. Gibson et al., 1955; J. J. Gibson, 1944). Over the decade of the 1950s, Gibson conducted a series of experiments that examined the effect of motion on the perception of object properties (e.g., J. J. Gibson & Gibson, 1957; J. J. Gibson, 1957), as well as the significance of the perceptual flow of visual stimulation during locomotion for the control of action (J. J. Gibson, 1958). Among the salient findings of these studies is that many object properties (e.g., slant, shape, relative

distance) are most readily perceived from a moving point of observation. Such results, of course, run contrary to what would be expected from the received approach to vision, with its legacy of perspectivism and its stance that perception is based on the reception of static visual images because movements of the perceiver (or of an object) should make veridical perception less likely owing to, for example, a blurring of the retinal image. Instead, Gibson found that action is typically a constituent of the act of veridical perceiving.

This body of work culminated in two books in which J. J. Gibson (1966, 1979) offers a radical re-conceptualization of perceiving processes and, concomitantly, of the stimulus information that is available to be perceived in the surround through perception-action. Recognizing that action is a vital facet of perceiving, Gibson proposed that vision is best considered as the function of a *perceptual system*. The visual perceptual system includes the eye and the visual pathways and brain, to be sure, but also the movements of the head, neck, and entire body—*movements that facilitate the detection of structure in a changing array of visual information*. In short, vision—or better, seeing—is a process of *perception-action*, not the mere passive reception of patterns of visual stimulation or images.

There is now abundant evidence in support of this perspective. Vision is critical in guiding self-directed action among animals occupying illuminated niches (Lee, 1976; Smart et al., 2020; Warren, 1994); and perceiving objects in motion or from a moving point of observation facilitates veridical visual perception (e.g., J. J. Gibson & Gibson, 1957; J. J. Gibson, 1957; Johansson, 1994; Kellman, 1984; Wallach & O'Connell, 1953).¹⁹

Further, as regards the stimulus information for perceiving, Gibson proposed a domain of *ecological optics* that considers the ways in which light is *structured* by surfaces in the environment, giving rise to an array of reflected light (a medium) surrounding a perceiver that can serve as source of potential information. Ecological optics, unlike physical optics, examines the ways in which light reflected from surfaces in the econiche is structured at a scale of analysis relative to a perceiver. This ambient array of reflected light is densely and hierarchically structured owing to the multifaceted character of the environment itself, and providing open-ended possibilities for perceptual learning and discovery for perceivers engaged in its exploration. In a manner perhaps envisioned by the Cubists, the perceiver moving with respect to an object, figure, landscape, or cityscape can perceive structure specific to them from multiple points of observation; and in the course of perceiving over time, structure that is invariant across a moving point of observation is revealed. Finally, because the structure in the array of reflected light, in principle, corresponds univocally to (i.e., specifies) the character of available objects and events in the surround, it raises the possibility that the latter are perceived directly—that is, without a perceiver needing to draw on nonperceptual constructive or inferential processes, for example, in the tradition of Helmholtz.

The influence of the 19th century advances in science and technology on Gibson's thinking are apparent. For example, whereas the received views of visual perception would seem mostly to tack on evolutionary considerations to their long-standing assumptions, Gibson *begins* by considering the function of vision for animate organisms from an evolutionary perspective. In doing so, he starts the last two books on perception in a highly unorthodox manner. Whereas texts on perception invariably begin with treatments of physics, sensory processes, and anatomy, Gibson begins by considering the nature of the environment that organisms (and terrestrial organism, at that) have adapted to over time. Surely, an evolutionary approach to perceiving should start there. In part for this reason, J. J. Gibson (1979) describes his treatment of perception as an ecological approach.²⁰

This is a field-theoretical approach to perception, where the perceptual field encompasses both the environment that is perceived and the perceiver's own body. His now widely cited and studied concept of affordance is prototypical of this stance (Gibson, 1979). Moreover, like field theory in physics, much of Gibson's work and the research it has inspired, challenges the classical, orthogonal dimensions of space and time. For example, Lee (1976) found that impending contact with a surface can be mathematically specified in the flow of visual information as a constant over time. This study then integrates space and time, representing in the domain of the control of action "a shift from thinking about behavior as guided by the perception of space, to thinking about behavior as a spatiotemporal coordination problem" (Warren, 2007, pp. 243–244). Likewise, evidence for perceiving shape and configuration by detecting structure that is invariant over time abandons the tendency in perceptual theory to maintain a categorical distinction between space and time.

Finally, as we have seen, 19th technologies such as trains and photography were instrumental in highlighting the temporal dimensions of experience, and in the former case, contributing eventually to the conceptual integration of space/time. Recall that some of Gibson's earliest research focused on the individual's experience of moving through the environment by car and by plane. He has traced the origins of his interest in perceiving over time to his early experiences as the son of a railroad worker, when as a child he observed the flow of the environment from the rear end of a moving train (Gibson, 1967). Gibson was not alone in drawing on such experiences when reflecting on character of immediate experience; so did William James whose writings later influenced Gibson (see, James, 1988, p. 220).

Gibson's thinking and his ecological approach in particular have typically been treated by mainstream theorists as an outlier, at best, in relation to contemporary perceptual theory. Though that attribution may be accurate as far as it goes, it may tell us less about Gibson's approach than it does about the relationship between mainstream perceptual theory and recent trends in the natural sciences (Turvey, 2019). The historical examination of theories of visual perception that we have undertaken here indicates that contemporary mainstream approaches remain wedded to assumptions about visual perception that arose in early periods of modern science around the 16th and 17th centuries and were informed by developments in the visual arts just before that era. Gibson's approach, although it appears to have taken shape independently of early 20th century developments in the visual arts, like them it drew inspiration from the cultural transformations generated by 19th century science and technology.

8 | CONCLUSIONS

Why should considerations of the history of the visual arts matter either for the study of visual perception or for the history of psychology? First, the historical analysis offered above indicates that developments in the visual arts *already have had* profound effects on the formulation of theories of visual perception, particularly as they concern the basis for "space perception." For this reason, revealing some of the historical roots of these received assumptions could well be beneficial for self-reflection in the ongoing development of visual perception theory.

Second, developments in the visual arts can be symptomatic of wider trends in the culture at large, and for this reason they may shed light historically on shifting ideas in psychological theory. And because visual artists and vision scientists both have a commitment to the character of visual experience, the relationships between these domains may be particularly illuminating (e.g., Vitz & Glimcher, 1983). As we have seen above, this was the case with the development of linear perspective, which reflects the Renaissance shift toward Humanism and its focus on an individual's experience. We have also seen the ways that early 20th century developments in art history reflected the 19th century cultural shift to dynamic, field-oriented conceptions of the natural world—a shift that only begins to be apparent in perceptual psychology with Gestalt theory and the emergence of ecological psychology (Heft, 2001).

At a minimum, developments in the arts can serve as a barometer of wider changes in the sciences and in the culture at large—and the history of the arts typically bears this out. Understandably, "normal science" in a Kuhnian sense can be rather inward-looking and insular because those tendencies can keep a domain of science well focused and, as a result, productive. But they can also constrain growth and leave a domain of inquiry out of step with broader trends in other sciences and the culture broadly considered. Visual scientists will be well served by being attuned to the effects of past historical influences on current thought for better and for worse, and by widening their scope to consider the broader cultural context within which the sciences operate.

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ENDNOTES

¹Stating that “it was put into practice” at that time is to recognize that there had been sporadic instances of at least rudimentary linear perspective at earlier times. For example, there is some indication of the use of perspective and other suggestions of depth on Greek vase paintings and Roman murals (Hyman, 2006), but by the Middle Ages it was abandoned likely due to its incompatibility with broader cultural concerns. King (2000) suggests there was a resistance in Medieval Christian art to depictions that appeared to be illusory rather than true to form. Through the development of systematic techniques beginning in the 14th century, linear perspective was “reinvented” and took hold with a shift in cultural perspective (see below).

²For a different view, see Elkins (1994).

³Some of these types of murals were uncovered in recent excavations at Pompeii.

⁴Gary Hatfield (personal communication, June 4, 2019) notes that there are Renaissance paintings that represent various moments but do so within the same painting by showing the protagonist at three different locations in the frame; but this was less common after artificial perspective was invented.

⁵Descartes' treatment of vision in the 1637 *Dioptrics* has at times been interpreted as positing rational processes that operate on the retinal image to give rise to a mental representation of three-dimensional space. Hatfield (2015) has recently argued that this is a misreading of Descartes because in keeping with his mechanistic approach to bodily processes, he offered an account that proposed how “we experience distance in a phenomenally immediate manner; that is, we find ourselves confronted by a visual world in which objects are phenomenally presented as being at various distances” (p. 167). In other words, Descartes proposes, on this reading, that distance can be experienced in a direct rather than cognitive mediated manner.

⁶In any case, a “live” retina is translucent, not opaque (Carello & Turvey, 2020).

⁷Owing to length considerations, I set aside here a discussion of the place of Kant's writings and those of his successors concerning space perception.

⁸In brief, Helmholtz proposed that each retinal element acquires as a result of prior movements a “local sign” that indexes its spatial location relative to the body (see, Hatfield, 1990, p. 174). Attributing “local signs” of spatial directions to retinal elements was much discussed in mid-19th century vision theory (e.g., Lotze, 1852, cited in Boring, 1929, p. 259).

⁹There is some historical irony with the word choice here, suggesting that what is perceived is a scene, not the world. Recall, as noted above, that Euclid's use of geometry to explain visual perception was initially applied to the scenography.

¹⁰Owing to space limitations I am unable to offer here a critical analysis of the assumption of the retinal image. Suffice it to say, that not long after Kepler's writings, Descartes in 1637 recognized that it is necessary to avoid this misstep: He asked even if there is such an image are there “yet other eyes in our brain with which we could perceive it”? (quoted in Hatfield, 2015, footnote 12). Other arguments can be marshaled to challenge this assumption (see, e.g., Carello & Turvey, 2020; Gibson, 1979, p. 60).

¹¹I thank Alan Costall for bringing this passage to my attention.

¹²To argue, as some might later be inclined to do, that experiencing “space” is attributable to a species' evolutionary history, that “perceiving space” is adaptive, simply begs James' questions (see Oyama, 1985).

¹³He describes Kant's a priori category of space as “mythological because I am conscious of no such Kantian machine-shop in my mind, and feel no call to disparage the powers of poor sensation in this way. I have no introspective experience of mentally producing or creating space. My space intuitions occur not in two times, but in one” (James, 1981, p. 905).

¹⁴Merleau-Ponty (1963) makes a similar argument (p. 255).

¹⁵In fact, there is a region of viewing angles within which the perspective painting retains its realistic-looking appearance (Wartofsky, 1980).

¹⁶“[I]n order to arrive at an isotropic space, the subject must leave his place, his [stationary] point of view on the world, and think himself into a sort of ubiquity” (Merleau-Ponty, 1963, p. 266)

¹⁷I set aside recent transformations in digitizing images, which of course undermines any notion of a geometric isomorphism between the field of view and photographic “image.”

¹⁸This last sentence can be found nearly verbatim in the phenomenologist Merleau-Ponty (1963) chapter on Space in *The Phenomenology of Perception* (p. 67), and in Gibson's *The Ecological Approach to Visual Perception* (pp. 198–199). In the case of both, perceiving is reconceptualized as an action of the body—a process of perception-action.

¹⁹Other examples include research showing that visual information generated through self-movement (i.e., optic flow) is utilized to guide the direction and speed of locomotion as well as serving as information specifying time-to-contact with a surface (Fajen, 2005; Lee, 1976); that movement facilitates the visual detection of depth at an edge (E. J. Gibson & Walk, 1960; Rader et al., 1980); that movement facilitates the visual judgment of surface height for sitting relative to the body size by adults (Mark, 1987), for passable aperture widths (Warren & Whang, 1987), and the size of gaps in ground surfaces (Mark et al., 1999). It also reduces the effect of classic three-dimensional visual illusions, such as the Ames Room, which are intended to be judged from a stationary position (e.g., Runeson, 1988).

²⁰Gibson also wrote about picture perception throughout his career, corresponded with the art historian Ernst Gombrich on several occasions, and published several papers in the journal *Leonardo*, which is dedicated to the relationships between the arts, the sciences, and technology

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REFERENCES

- Apollinaire, G., & Emert, D. (2010). *Cubism*. Parkstone Press.
- Arheim, R. (1954). *Art and visual perception: A psychology of the creative eye*. University of California Press.
- Berger, J. (1972). *Ways of seeing*. British Broadcasting Corporation.
- Berkeley, G. (1709). *An essay towards a new theory of vision and other writings*. Dutton.
- Berkeley, G. (1733). *The theory of vision vindicated and explained*. Tonson.
- Bjork, D. W. (1983). *The compromised scientist: William James in the development of American psychology*. Columbia University Press.
- Boring, E. G. (1929). *A history of experimental psychology*. Century.
- Boring, E. G. (1942). *Sensation and perception in the history of experimental psychology*. Appleton-Century.
- Carello, C., & Turvey, M. T. (2020). Challenging the axioms of perception: The retinal image and the visibility of light. In J. B. Wagman, & J. C. Blau (Eds.), *Perception as information detection: Reflections on Gibson's ecological approach to visual perception* (pp. 51–69). Routledge.
- Cassirer, E. (1963). *The individual and the cosmos in Renaissance philosophy*. (Translated by Mario Domani) Angelico Press (Originally published in 1927).
- Clark, S. (2007). *Vanities of the eye: Vision in early modern European culture*. Oxford University Press.
- Cornsweet, T. (2017). *Seeing: How light tells us about the world*. University of California Press.
- Costall, A. (1993). Beyond linear perspective: A cubist manifesto for visual science. *Image and Vision Computing*, 11, 334–341.
- Daniel, M. R. (1998). *Edgar Degas, photographer*. Metropolitan Museum of Art.
- Edgerton, S. Y. (2009). *The mirror, the window, and the telescope: How Renaissance linear perspective changes our vision of the universe*. Cornell University Press.
- Elkins, J. (1994). *The poetics of perspective*. Cornell University Press.
- Fajen, B. R. (2005). Calibration, information, and control strategies for braking to avoid collision. *Journal of Experimental Psychology: Human Perception and Performance*, 31, 480–501.
- Friedberg, A. (2006). *The virtual window: From Alberti to Microsoft*. MIT Press.
- Fry, E. F. (1966). *Cubism*. Thames and Hudson.
- Galisson, P. (2003). *Einstein's clocks, Poincare's maps: Empires of time*. W.W. Norton.
- Gernsheim, H. (1986). *A concise history of photography* (3rd ed.). Dover.
- Gibson, E. J., & Walk, R. (1960). The "visual cliff". *Scientific American*, 202, 64–71. <https://doi.org/10.1038/scientificamerican0460-64>
- Gibson, J. J. (1944). History, organization, and research activities of the psychological test film unit. *Army Air Forces. Psychological Bulletin*, 38, 432–468.
- Gibson, J. J. (1957). Optical motions and transformations as stimuli for visual perception. *Psychological Review*, 64, 288–295. <https://doi.org/10.1037/h0044277>
- Gibson, J. J. (1958). Visually controlled locomotion and visual orientation in animals. *British Journal of Psychology*, 49, 182–194. https://doi.org/10.1207/s15326969eco103%264_2
- Gibson, J. J. (1966). *The senses considered as perceptual systems*. Houghton-Mifflin.

- Gibson, J. J. (1967). James J. Gibson. In E. G. Boring, & G. Lindsey (Eds.), *A history of psychology in autobiography* (Vol. 5). Appleton-Century Crofts.
- Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton-Mifflin.
- Gibson, J. J., & Crooks, L. E. (1938). A theoretical field analysis of automobile-driving. *American Journal of Psychology*, 51, 453–471.
- Gibson, J. J., & Gibson, E. J. (1957). Continuous perspective transformations and the perception of rigid motion. *Journal of Experimental Psychology*, 54, 129–138. <https://doi.org/10.1037/h0041890>
- Gibson, J. J., Olum, P., & Rosenblatt, F. (1955). Parallax and perspective during aircraft landings. *American Journal of Psychology*, 68, 372–385.
- Gideon, S. (1944). *Space, time, and architecture*. Harvard University Press.
- Gleizes, A., & Metzinger, J. (1912). *Du Cubisme*. Eugène Figuère Editeurs. https://monoskop.org/images/7/73/Gleizes_Metzinger_1912_1965_Cubism.pdf
- Glennerster, A. (2007). Marr's vision: Twenty-five years on. *Current Biology*, 17(11), R397–R399. <https://doi.org/10.1016/j.cub.2007.03.035>
- Gombrich, E. H. (1960). *Art and illusion: A study in the psychology of pictorial representation*. Princeton University Press.
- Hagen, M. A. (1980). Generative theory: A perception theory of pictorial representation. In M. A. Hagen (Ed.), *The perception of pictures* (Vol. 2, pp. 3–46). Academic Press.
- Hamlyn, D. W. (1961). *Sensation and perception: A history of philosophy of perception*. Routledge & Kegan Paul.
- Hatfield, G. (1990). *The natural and the normative: Theories of spatial perception from Kant to Helmholtz*. MIT Press.
- Hatfield, G., & Epstein, W. (1979). The sensory core and the medieval foundations of early modern perceptual theory. *Isis*, 70, 363–384. <https://doi.org/10.1086/352281>
- Hatfield, G. (2015). On natural geometry and seeing distance directly in Descartes. In V. de Risi (Ed.), *Mathematizing space: The objects of geometry from antiquity to the early modern age* (pp. 157–191). Birkhäuser.
- Heft, H. (2001). *Ecological psychology in context: James Gibson, Roger Barker, and the legacy of William James's radical empiricism*. Lawrence Erlbaum Associates, Publishers.
- Heft, H. (2003). Affordances, dynamic experience, and the challenge of reification. *Ecological Psychology*, 15, 149–180. https://doi.org/10.1207/S15326969ECO1502_4
- Heft, H. (2020). Revisiting “The discovery of the occluding edge and its implications for perception” 40 years on. In J. B. Wagman, & J. J. C. Blau (Eds.), *Perception as information detection: Reflections on Gibson's ecological approach to visual perception* (pp. 188–204). Routledge.
- Helmholtz, H. (1924–1925). In Southall, J. P. C. Translated by, *Handbook of physiological optics*. Optical Society of America.
- Hoffmann, B. (1983). *Relativity and its roots*. New York: Scientific American.
- Hyman, J. (2006). *The objective eye: Color, form, and reality in the theory of art*. The University of Chicago Press.
- James, W. (1920). Remarks on Spencer's definition of mind as correspondence. In *Collected essays and reviews* (pp. 43–68). Longmans, Green (Original work published in 1878).
- James, W. (1976). *Essays in radical empiricism*. Harvard University Press (Original work published 1912).
- James, W. (1981). *The principles of psychology*. Harvard University Press (Original work published 1890).
- James, W. (1988). *Manuscript lectures*. Harvard University Press (Original notes written in 1895–1896).
- James, W. (1996). *A pluralistic universe*. University of Nebraska Press (Original work published in 1909).
- Jennings, H. S. (1906). *Behavior of the lower organisms*. Columbia University Press.
- Johansson, G. (1994). Configurations in event perception. In G. Jansson, S. S. Bergström, & W. Epstein (Eds.), *Perceiving events and objects* (pp. 29–122). Lawrence Erlbaum.
- Joyce, P. (1988). *Hockney on photography*. Harmony Books.
- Kellman, P. (1984). Perception of three-dimensional form by human infants. *Perception and Psychophysics*, 36, 353–358. <https://doi.org/10.3758/BF03202789>
- King, R. (2000). *Brunelleschi's Dome: How a Renaissance genius reinvented architecture*. Penguin.
- LaPorte, P. M. (1949). Cubism and science. *The Journal of Aesthetics and Art Criticism*, 7(3), 243–256.
- Lee, D. (1976). A theory of visual control of braking based on information about time-to-collision. *Perception*, 5(4), 437–459. <https://doi.org/10.1068/p050437>
- Leger, F. (1914). Contemporary achievements in painting. In Fry, E. F. (1966), *Cubism* (pp. 135–139). Thames and Hudson.
- Lindberg, D. C. (1976). *Theories of vision from Al-Kindi to Kepler*. University of Chicago Press.
- Mark, L. (1987). Eyeheight-scaled information about affordances: A study of sitting and stair climbing. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 361–370. <https://doi.org/10.1037/0096-1523.13.3.361>
- Mark, L. S., Jiang, Y., King, S. S., & Paasche, J. (1999). The impact of visual exploration on judgments of whether a gap is crossable. *Journal of Experimental Psychology: Human Perception and Performance*, 25, 287–295. <https://doi.org/10.1037/0096-1523.25.1.287>

- Marr, D. (1982). *Vision: A computational investigation into the human representation and processing of visual information*. W.H. Freeman.
- Merleau-Ponty, M. (1963). *The phenomenology of perception*. (C. Smith, transl.). Routledge & Kegan Paul (Originally published in 1945).
- Metzinger, J. (1910). Note on painting. In Fry, E. F. (1966), *Cubism* (pp. 59–61). Thames and Hudson.
- Miller, A. I. (2001). *Einstein, Picasso: Space, time, and the beauty that causes havoc*. Basic Books.
- Mook, D. E., & Vargish, T. (1987). *Inside relativity*. Princeton University Press.
- Oyama, S. (1985). *The ontogeny of information: Developmental systems and evolution*. Cambridge University Press.
- Panofsky, E. (1991). *Perspective as symbolic form*. Zone Books (Originally published in 1927).
- Pastore, N. (1971). *Selective history of theories of visual perception: 1650–1950*. Oxford University Press.
- Rader, N., Bausano, M., & Richards, J. E. (1980). On the nature of the visual-cliff avoidance response in human infants. *Child Development*, 51, 61–68. <https://doi.org/10.1111/j.1467-8624.1980.tb02509.x>
- Richardson, J. (1991). *A life with Picasso* (Vol. I). Random House.
- Richardson, R. D. (2006). *William James: In the maelstrom of American modernism*. Houghton Mifflin.
- Rock, I. (1985). *The logic of perception*. MIT Press.
- Runeson, S. (1988). The distorted room illusion, equivalent configurations, and the specificity of static optic arrays. *Journal of Experimental Psychology: Human Perception and Performance*, 14, 295–304. <https://doi.org/10.1037//0096-1523.14.2.295>
- Schivelbusch, W. (1986). *The railway journey: The industrialization of time and space in the 19th century*. The University of California Press.
- Sedgwick, H. A. (1980). The geometry of spatial layout. In M. A. Hagen (Ed.), *The perception of pictures* (Vol. 1, pp. 33–90). Academic Press.
- Sheets-Johnstone, M. (2016). *Insides and outsides: Interdisciplinary perspectives on animate nature*. Imprint Academic.
- Smith, A. M. (2015). *From sight to light: The passage from ancient to modern optics*, The University of Chicago Press.
- Smart, L. J., Hassebrock, J. A., & Teaford, M. A. (2020). Acting is perceiving: Experiments on perception of motion in the world and movements of the self, an update. J. B. Wagman & J. J. C. Blau *Perception as information detection: Reflections on Gibson's ecological approach to visual perception*, (174–187). Routledge.
- Snyder, J. (1980). Picturing vision. *Critical Inquiry*, 6, 499–526. <https://doi.org/10.1086/448062>
- Stoffregen, T. A. (2013). On the physical origins of inverted optic images. *Ecological Psychology*, 25, 369–382.
- Straker, S. (1981). Kepler, Tycho, and the 'Optical Part of Astronomy': The genesis of Kepler's theory of pinhole images. *Archive for History of Exact Sciences*, 24, 267–293.
- Turvey, M. T. (2019). *Lectures on perception: An ecological perspective*, Routledge.
- Vargish, D. E., & Mook, T. (1991). *Inside relativity*. Princeton University Press.
- Vitz, P. C., & Glimcher, A. B. (1983). *Modern art and modern science: The parallel analysis of vision*. Praeger.
- Wallach, H., & O'Connell, D. N. (1953). The kinetic depth effect. *Journal of Experimental Psychology*, 45(4), 205–217. <https://doi.org/10.1037/h0056880>
- Warren, W. H. (1994). Visually controlled locomotion: 450 years later. *Ecological Psychology*, 10, 177–219.
- Warren, W. H. (2007). Action-scaled information for the visual control of locomotion. In G. Pepping, & M. A. Greal (Eds.), *Closing the gap: The scientific writings of David N. Lee* (pp. 243–258). Lawrence Erlbaum.
- Warren, W. H., & Whang, S. (1987). Visual guidance of walking through apertures: Body-scaled information for affordances. *Journal of Experimental Psychology: Human Perception and Performance*, 13, 371–383. <https://doi.org/10.1037//0096-1523.13.3.371>
- Wartofsky, M. W. (1980). Visual scenarios: The role of representation in visual perception. In M. A. Hagen (Ed.), *The perception of pictures* (Vol. 2, pp. 131–151). Academic Press.
- Wartofsky, M. W. (1973). Perception, representation, and the forms of action: Towards an historical epistemology. In M. W. Wartofsky (Ed.), *Models: Representation and the scientific understanding*. *Boston Studies in the Philosophy of Science* (Vol. XLVIII, pp. 188–210). D. Reidel Publishing Co.
- Wofflin, H. (2015). *Principles of art history: The problem of the development of style in early modern art*. Getty Research Institute (Originally published in 1915).
- Woodworth, R. S. (1921). *Psychology: A study of mental life*. Henry Holt.

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